Divergence Theorem

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Application of Gauss's Law: Differential volume element

- We shall now obtain an exact relationship, by allowing the volume element Δv to shrink to zero.

$$\left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)=\frac{\oint_S \mathbf{D}\cdot d\mathbf{S}}{\Delta v}=\frac{Q}{\Delta v}$$

$$\left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)=\lim_{\Delta v\to 0}\frac{\oint_S \mathbf{D}\cdot d\mathbf{S}}{\Delta v}=\lim_{\Delta v\to 0}\frac{Q}{\Delta v}$$

- The last term is the volume charge density ρ_v, so that:

$$\left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)=\lim_{\Delta v\to 0}\frac{\oint_S \mathbf{D}\cdot d\mathbf{S}}{\Delta v}=\rho_v$$

Application of Gauss's Law: Differential volume element

- Let us now consider one information that can be obtained from the last equation:

$$\left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)=\lim_{\Delta v\to 0}\frac{\oint_S \mathbf{D}\cdot d\mathbf{S}}{\Delta v}$$

- This equation is valid not only for electric flux density D, but also to any vector field A to find the surface integral for a small closed surface.

$$\left(\frac{\partial A_x}{\partial x}+\frac{\partial A_y}{\partial y}+\frac{\partial A_z}{\partial z}\right)=\lim_{\Delta v\to 0}\frac{\oint_S \mathbf{A}\cdot d\mathbf{S}}{\Delta v}$$

- This operation received a descriptive name, *divergence*. The divergence of A is defined as:

$$\text{Divergence of } \mathbf{A} = \text{div } \mathbf{A} = \lim_{\Delta v\to 0}\frac{\oint_S \mathbf{A}\cdot d\mathbf{S}}{\Delta v}$$

Divergence

$\nabla \cdot \mathbf{D}$

$$\text{div}\,\mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

Rectangular

$\nabla \cdot \mathbf{D}$

$$\text{div}\,\mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

Cylindrical

$\nabla \cdot \mathbf{D}$

$$\text{div}\,\mathbf{D} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$

Spherical

Divergence

- **Example**
  If $D = e^{-x}\sin y\, a_x - e^{-x}\cos y\, a_y + 2z\, a_z$, find div D at the origin and $P(1,2,3)$.

$$\text{div}\,\mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = -e^{-x}\sin y + e^{-x}\sin y + 2 = \underline{\underline{2}}$$

Regardless of location the divergence of D equals 2 C/m^3.

The Vector Operator ∇ and The Divergence Theorem

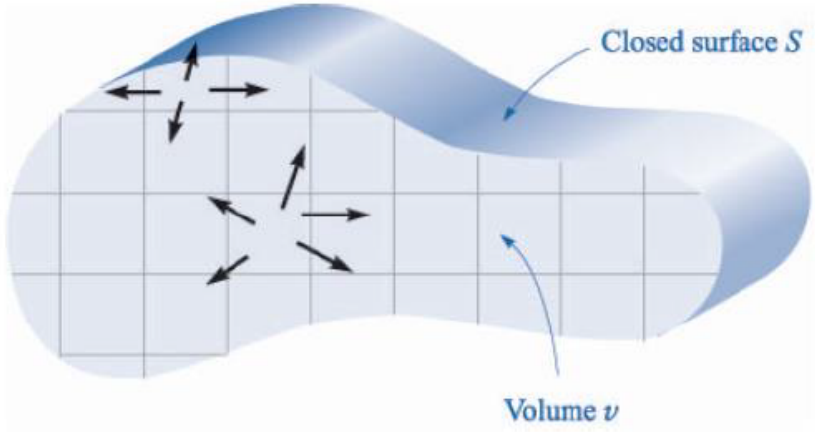
- We shall now give name to a theorem that we actually have obtained, the Divergence Theorem:

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv = \int_{\text{vol}} \nabla\cdot\mathbf{D} dv$$

- The first and last terms constitute the divergence theorem:

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{vol}} \nabla\cdot\mathbf{D}\, dv$$

"The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed by the closed surface."



Maxwell's First Equation (Electrostatics)

- We may now rewrite the expressions developed until now:

$$\text{div}\,\mathbf{D} = \lim_{\Delta v \to 0} \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{\Delta v}$$

$$\text{div}\,\mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\nabla \cdot \mathbf{D} = \rho_v$$

Maxwell's First Equation
Point Form of Gauss's Law

- This first of Maxwell's four equations applies to electrostatics and steady magnetic field.
- Physically it states that the electric flux per unit volume leaving a vanishingly small volume unit is exactly equal to the volume charge density there.

■ **Example**

Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\,\mathbf{a}_x + x^2\,\mathbf{a}_y$ C/m^2 and the rectangular parallelepiped formed by the planes $x = 0$ and 1, $y = 0$ and 2, and $z = 0$ and 3.

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv$$ Divergence Theorem

$$\nabla\cdot\mathbf{D} = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(x^2) = 2y$$

$$\oint_S \mathbf{D}_S\cdot d\mathbf{S} = \int_0^3\int_0^2 (\mathbf{D})_{x=0}\cdot(-dydz\ \mathbf{a}_x) + \int_0^3\int_0^2 (\mathbf{D})_{x=1}\cdot(dydz\ \mathbf{a}_x)$$
$$+ \int_0^3\int_0^1 (\mathbf{D})_{y=0}\cdot(-dxdz\ \mathbf{a}_y) + \int_0^3\int_0^1 (\mathbf{D})_{y=2}\cdot(dxdz\ \mathbf{a}_y)$$

$$\int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv = \int_{z=0}^3\int_{y=0}^2\int_{x=0}^1 (2y)dxdydz$$
$$= x\Big|_0^1\, y^2\Big|_0^2\, z\Big|_0^3$$
$$= \underline{\underline{12\ \text{C}}}$$

But $(D_x)_{x=0} = 0,\ \ (D_y)_{y=0} = (D_y)_{y=2}$

$$\oint_S \mathbf{D}_S\cdot d\mathbf{S} = \int_0^3\int_0^2 (D_x)_{x=1}dydz = \int_0^3\int_0^2 2ydydz = \underline{\underline{12\ \text{C}}}$$

$$\therefore \oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv = 12\ \text{C}$$

3.22 Let $\mathbf{D} = 8\rho \sin\phi \mathbf{a}_\rho + 4\rho\cos\phi \mathbf{a}_\phi \text{C/m}^2$. (*a*) Find div **D**. (*b*) Find the volume charge density at $P(2.6, 38°, -6.1)$. (*c*) How much charge is located inside the region defined by $0 < \rho < 1.8$, $20° < \phi < 70°$, $2.4 < z < 3.1$?

Let $\mathbf{D} = 8\rho \sin\phi \, \mathbf{a}_\rho + 4\rho \cos\phi \, \mathbf{a}_\phi \text{ C/m}^2$.

a) Find div **D**: Using the divergence formula for cylindrical coordinates , we find $\nabla \cdot \mathbf{D} = \underline{12 \sin\phi}$.

$$\text{div } \mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial\phi} + \frac{\partial D_z}{\partial z}$$

b) Find the volume charge density at $P(2.6, 38°, -6.1)$: Since $\rho_v = \nabla \cdot \mathbf{D}$, we evaluate the result of part *a* at this point to find $\rho_{vP} = 12 \sin 38° = \underline{7.39 \text{ C/m}^3}$.

c) How much charge is located inside the region defined by $0 < \rho < 1.8$, $20° < \phi < 70°$, $2.4 < z < 3.1$? We use

$$Q = \int_{vol} \rho_v dv = \int_{2.4}^{3.1}\int_{20°}^{70°}\int_{0}^{1.8} 12 \sin\phi \rho \, d\rho \, d\phi \, dz = -(3.1 - 2.4)12\cos\phi\Big|_{20°}^{70°} \frac{\rho^2}{2}\Big|_0^{1.8}$$

$$= \underline{8.13 \text{ C}}$$

D3.7. In each of the following parts, find a numerical value for div $\mathbf{D}$ at the point specified: (*a*) $\mathbf{D} = (2xyz - y^2)\mathbf{a}_x + (x^2z - 2xy)\mathbf{a}_y + x^2y\mathbf{a}_z$ C/m^2 at $P_A(2, 3, -1)$; (*b*) $\mathbf{D} = 2\rho z^2 \sin^2\phi\,\mathbf{a}_\rho + \rho z^2 \sin 2\phi\,\mathbf{a}_\phi + 2\rho^2 z \sin^2\phi\,\mathbf{a}_z$ C/m^2 at $P_B(\rho = 2, \phi = 110°, z = -1)$; (*c*) $\mathbf{D} = 2r \sin\theta \cos\phi\,\mathbf{a}_r + r\cos\theta\cos\phi\,\mathbf{a}_\theta - r\sin\phi\,\mathbf{a}_\phi$ C/m^2 at $P_C(r = 1.5, \theta = 30°, \phi = 50°)$.

Ans. (*a*) −10.00; (*b*) 9.06; (*c*) 1.29

Divergence

$$\text{div}\,\mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$ Rectangular

$$\text{div}\,\mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$ Cylindrical

$$\text{div}\,\mathbf{D} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 D_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta D_\theta) + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$ Spherical

D3.9. Given the field $\mathbf{D} = 6\rho \sin\frac{1}{2}\phi\, \mathbf{a}_\rho + 1.5\rho \cos\frac{1}{2}\phi\, \mathbf{a}_\phi$ C/m^2, evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\phi = 0$, $\phi = \pi$, $z = 0$, and $z = 5$.

Ans. 225; 225

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$


Thank
you